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LEADING ARTICLE

Missing Data in Sport Science: A Didactic Example Using Wearables in American Football

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Abstract

Data are often recorded from athletes to make decisions regarding the mitigation of injuries or the enhancement of performance. However, data collection in the real-world is difficult, and it is common for data to be missing from a particular training session due to equipment malfunction, athlete non-compliance, etc. The statistical community has long recognized that proper handling of missing data is vital to unbiased analyses and decision making, yet most dashboards in sport science and medicine do not recognize the issues introduced by missing data and practitioners are largely unaware that their displays are conveying biased information. The goal of this leading article is to show how real-world data can violate the 'missing completely at random' assumption in an American Football example and then demonstrate some potential imputation solutions which appear to maintain the underlying properties of the data in the presence of missingness. Whether data are aggregated on a dashboard as simple histograms and averages or with higher-level analytics, violating the 'missing completely at random' assumption results in a biased dashboard. Practitioners need to insist that dashboard developers perform missing data analyses and impute data as needed so valid data-driven decisions can be made.

Key Points

- Real-world data in sport science and medicine often have missing entries and this missing data can bias interpretation unless handled carefully
- This example of wearable usage in American Football players shows that simply deleting missing data would bias any resulting analyses or interpretations
- The type of data missingness should continually be assessed and appropriate imputation considered in dashboards so clinicians can make valid interpretations in the field

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1 Introduction

It is widely recognized in the analytical community that missing data pose a potential hazard to valid interpretations, whether the goals are description, inference or prediction [1–3]. As practitioners and researchers, what we need to know is: can we proceed with an analysis or interpretation where we ignore the missingness of data or do we require some sort of analytical intervention in order to make valid descriptions, inferences or predictions [4]?

In 1976, Rubin classified missing data into three types: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) [5]. The data meet the MCAR assumption if the causes of the missing data are completely unrelated to the data itself. If the missing data are truly MCAR, we can delete observations with missing data without biasing the results (pairwise or listwise deletion, most commonly). However, the MCAR assumption often doesn't hold for real-world scenarios; for example, athletes may tire of completing surveys over time or sustain an injury mid-season and have increased amount of missing data late in the season. Data are considered MAR when they are missing systematically but are related to observed data points [6]. If the missing data are MAR, we can build a model for whether data is missing based upon

data we have collected such as sex, age, or time period in the study. Data are MNAR when data are missing systematically but related to events or factors which were not measured by the researcher or clinician. Both MAR and MNAR result in a biased analysis/interpretation if a deletion of missing data points occurs [7]. Most modern methods for handling missing data start with the MAR assumption, and so the focus of this manuscript will be on contrasting MCAR and MAR data handling.

Most commercial dashboards used in sport science implicitly assume that all data are MCAR. Off-the-shelf athlete management systems have no way of knowing that data 'should be there' and is missing if it is receiving external GPS monitoring data or data from a team's daily wearable. These dashboards display metrics such as individual athlete data points across time, group averages or distributions, and 7-day moving averages. In nearly all cases, the default method for handling a missing data point is deletion, which is also the method used in legacy enterprise software such as Microsoft Excel. Deletion is appropriate if the missing data are in fact MCAR, but may produce biased results if the MCAR assumption is violated. To avoid such biases, appropriate imputation methods need to be applied to sport science dashboards. Given that sport science dashboards are meant to be used in real-time to understand athlete and team progress daily, such missing data imputation methods need to be applicable in real-time. This manuscript presents a case study of wearable data missingness in the West Virginia University Football program for the 2021 season demonstrating established methods for assessing the MCAR assumption and some potential fixes that can be applied for valid real-time interpretations of wearable data during an athletic season. The goals of the paper are to raise awareness of the issue of missing data, to illustrate how different imputation methods can be applied to a dashboard using real-world data, and to discuss potential advantages and disadvantages of various imputation approaches.

2 Methods

During the 2021 Football season, 109 athletes were given Oura Ring wearables [Ōura Health Oy, Finland] to measure sleep quantity. Of these 109 athletes, 75 had at least 66% of their data recorded and are used in the present analysis. All athletes were college-age males (exact ages were not recorded). All data were recorded between August 7, 2021 and November 27, 2021. Athletes were instructed to wear their Oura Ring every night and synchronize their ring with the Oura application each morning. Periodic reminders were provided to the athletes to adhere to these criteria by the strength and conditioning staff. Oura Ring data was directly pulled into our database system via the Oura API.

2.1 Assessing MCAR Assumption and Missing Data Model

The basic data available for analysis in this study were athlete name, day number of the season, day of the week, athlete's position, and total sleep for that night. The MCAR assumption was directly examined using Little's Test, which implements a chi-squared test for whether a significant difference exists between the means of different missing-value patterns. In this test, a p-value < 0.05 is typically interpreted as evidence against the MCAR assumption [8]. A basic missing data model was also examined. This consists of dummy-coding missing data with a '0' and non-missing data with a '1' and using a multilevel logistic regression model where there are random intercepts for each athlete and a fixed effect for day number of the season, day of the week, and athlete's position. To aide in model convergence, day number of the season was z-scored prior to analysis but all results are transformed back to the original units for reporting. There were a total of 7188 non-missing data points and 1212 missing data points for the binary coded logistic regression. An analysis of variance can then be performed on the logistic regression and post-hoc comparisons as appropriate with Tukey's adjustment for multiplicity. The outcome of both Little's test and the missing data model will provide evidence for or against an MCAR assumption as well as indicate the pattern of missingness.

2.2 Imputation Methods and Evaluation

In recognition that various types of time series imputation might be appropriate in different circumstances for display and use on a dashboard in sport science, 11 different imputation methods were implemented to contrast against the standard-bearer method, which is pairwise deletion (i.e., deleting cases only when they are missing data necessary for a particular analysis). The univariate time series imputation methods were mean imputation, median imputation, last-observation-carried-forward imputation, random number imputation, simple moving average imputation, linear weighted moving average, exponentially weighted moving average, linear interpolation, spline interpolation, and Stineman interpolation. Multilevel model multiple imputation was also implemented using athlete position, day of the week and day number of the season as predictors using the methods set out by Schafer and Yucel [9]. All imputation methods were performed with the 'imputeTS' and 'mitml' R packages using the default settings. Across all athletes, only the 1212 missing

data points were imputed (i.e., missing data was not artificially simulated).

In recognizing that the goal of sport science analysts is to interpret data in real time to make clinical decisions, it was necessary to simulate what real-time imputations would look like in this population. To this end, starting 7 days into the season, a new imputation was performed every day for the entire data series up to that point and the data point reported for that athlete is what the sport scientist or coach would have seen on that day. For example, a simple imputation method such as mean imputation renders different imputations across time as data is iteratively added and the mean of the series changes. The one imputation type that would not vary across time is last-observation-carried-forward as this is simply the act of taking the last data point in a time series and 'filling down' any following missing data points until a new observed data point occurs.

Since we do not know the exact data generating mechanism behind the missing data, the goal of this analysis was not to 'prove' that any imputation method is inherently superior, but rather to quantitatively assess the internal consistency of each imputation algorithm and qualitatively contrast these methods against pairwise deletion. To examine internal consistency, the complete season's imputed data was used as the 'gold standard' and the mean absolute error (MAE) from the complete season was calculated for each day's sequential time series. In other words, each 'on the fly' imputation type is compared to it's own end-of-season imputation and the imputation types are not compared against each other. This renders 104 days of MAE readings which could then be plotted and examined for error bias (i.e., intercept of linear regression) and consistency across time (i.e., slope of linear regression). Ideally, we would like an imputation method that does not vary wildly across time and has a low MAE slope. In order to contrast the imputed time series data against deletion, we also graphically displayed individual athletes' 7-day moving averages across the season. This was selected because the 7-day moving average with missing data deletion is the standard reporting for Oura and many dashboards in sport science. This graphically represents the bias that may be introduced to practitioner interpretation if data are not MCAR and deletion of missing data is performed.

All analyses were performed in the R programming language [10] using the following packages: dplyr [11], lubridate [12], tidyr [13], lme4 [14], emmeans [15], ggplot2 [16], naniar [17], imputeTS [18], foreach [19], doParallel [20], zoo [21], caret [22], mitml [23], purrr [24], car [25], and ggpubr [26].

3 Results

3.1 MCAR Assumption and Missing Data Model

Little's test indicated that the data likely violated the MCAR assumption ($p < 0.001$). Furthermore, all three fixed effects in the missing data model were significant predictors of missingness ($p < 0.01$; Figs. 1, 2 and 3). Response rate significantly decreased across time (Fig. 1). Response rate varied by player position (Fig. 2), though post-hoc comparisons indicated that only the cornerback vs tight end/full back comparison was statistically different at the 0.05 alpha level. The response rates were profoundly different when stratified by day of the week (Fig. 3) with statistical differences between Saturday/Sunday and all

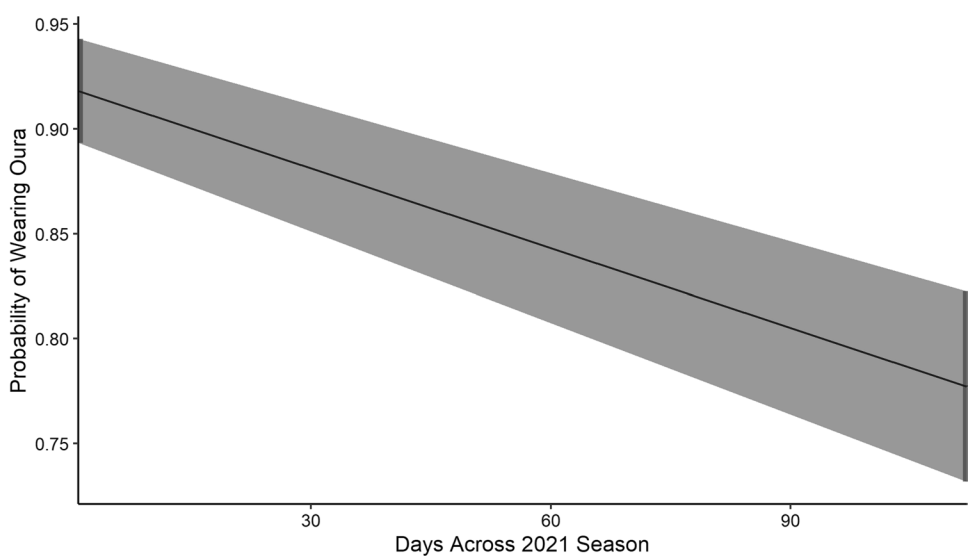

Fig. 1 Logistic regression estimate of probability of wearing Oura Ring across the season (*gray band* is 95% confidence interval)

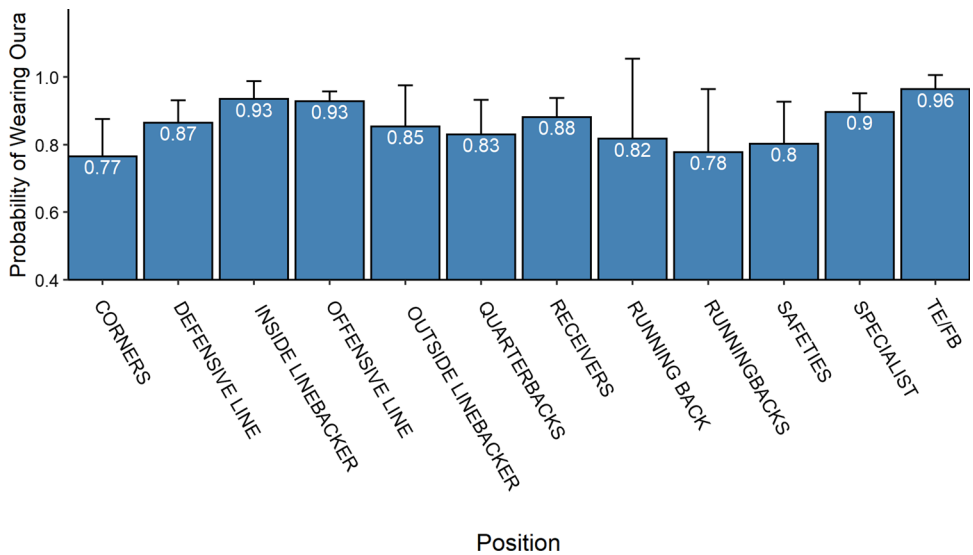

Fig. 2 Logistic regression estimate of probability of wearing Oura Ring stratified by player position. *White numbers* are the point estimate with *error bars* representing 95% confidence intervals

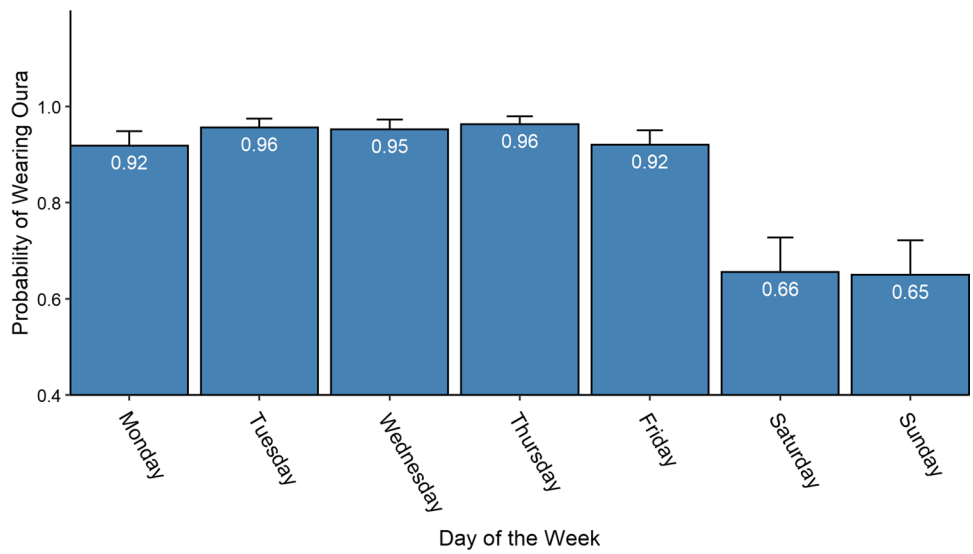

Fig. 3 Logistic regression estimate of probability of wearing Oura Ring stratified by day of the week. *White numbers* are the point estimate with *error bars* representing 95% confidence intervals

other time points as well as differences between Monday/Friday and Tuesday/Wednesday/Thursday ($p < 0.05$).

3.2 Imputation Evaluation

The MAE from the complete season data for each of the 11 imputation algorithms can be seen in Fig. 4. While the last-observation-carried-forward method does not vary across time (as noted in the Methods), it may have other negative properties (discussed below) that make it unsuitable for real-world use [3]. All of the moving average type imputation algorithms had a low bias and slope, suggesting strong internal consistency when computed serially. The multiple imputation method appeared to become more stable as data points were added, becoming wide-sense stable by Day 50.

Example plots of selected athletes clearly demonstrate the largest outlier of individual athlete data is the standard 7-day moving average which contains no imputation and instead does pairwise deletion. The random imputation method seems to deviate from the other imputation methods, but many of the imputation methods largely cluster together (see example athletes in Fig. 5).

4 Discussion

Using real-world data, this illustrates that the MCAR assumption is potentially violated for wearables in athletic environments and that applying any number of imputation methods to athletic dashboard data will likely result in more accurate conclusions than simply deleting missing data. This imputation is designed to maintain the underlying characteristics of the data compared with the status quo practice, pairwise deletion. Therefore, the current suggested

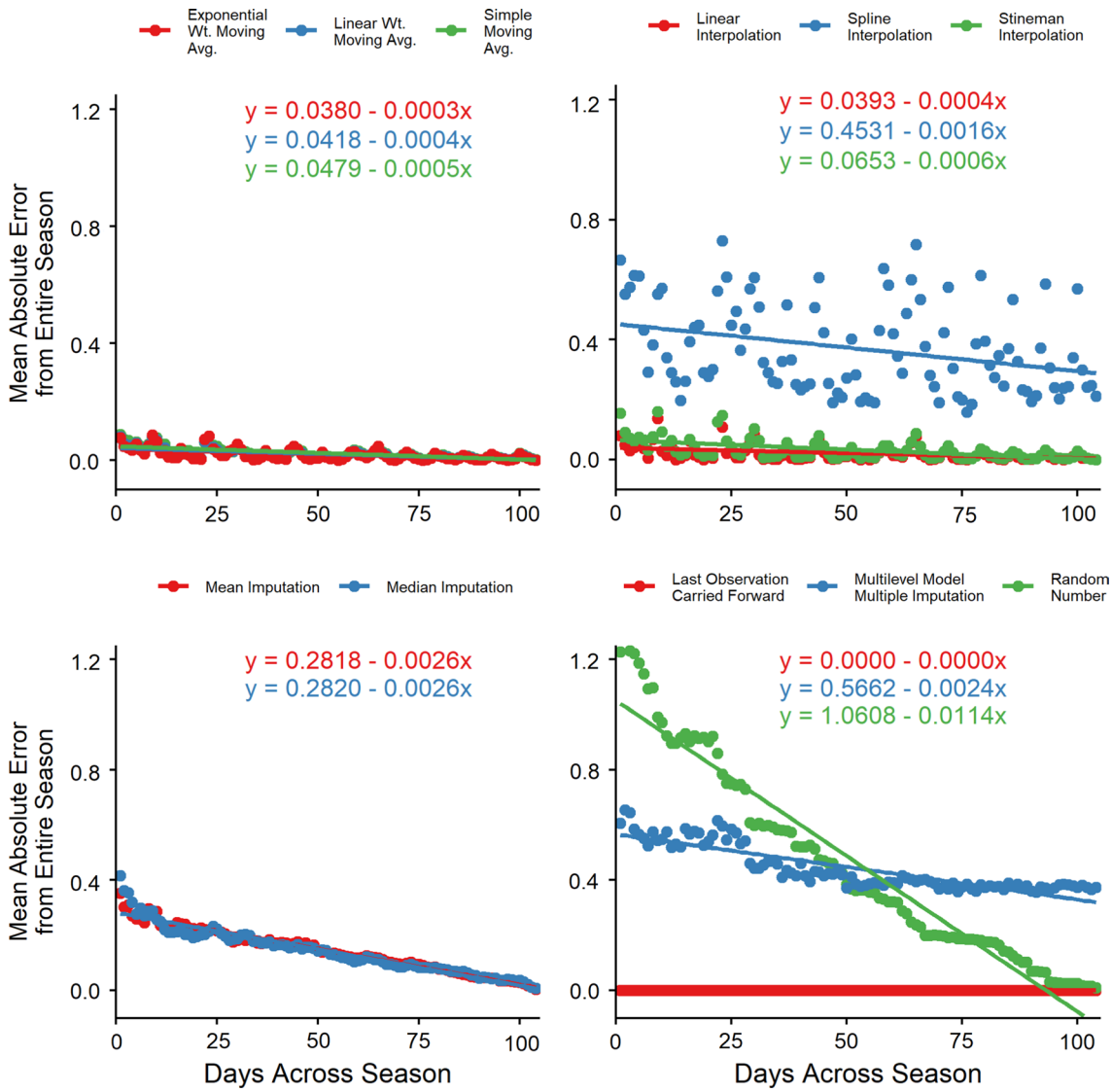

Fig. 4 For each imputation method, the mean absolute error from the final imputation across the entire season is compared with what the imputation would be on the exact day that a practitioner would see it. For each sub-plot, the color of the data points corresponds to the imputation method and the associated regression line and equation

course of action for dashboards in sport science is to not assume that data are MCAR unless proven otherwise and also consider an appropriate univariate imputation technique (such as exponentially weighted moving averages) or multiple imputation when possible.

The practical implications are hard to overstate. For example, if you had a day-of-the-week missingness pattern similar to what you see in Fig. 3 and you were interested in total sleep for an American Football team which plays nearly all of its games on Saturday or Sunday, you are not capturing much of the data before or after a game. The data you are capturing is potentially for players that are not traveling and would not experience the same sleep disturbances as your team overall. One may errantly conclude that sleeping patterns for your team are effective and not disturbed, something that is highly unlikely for a team traveling multiple time zones and arriving back from an 'away game' at 4 a.m. the night of a game. Even if data were filtered by 'athletes who traveled', the missingness pattern is still problematic because the reason for missing data or 'mechanism of missingness' is unknown. It is plausible that your highly compliant athletes are also the ones who are more likely to listen to their bodies or your non-compliant athletes are non-compliant due to discontent with their playing time; in either event, the pattern of missingness is important to understand and handle appropriately so that valid conclusions can be made.

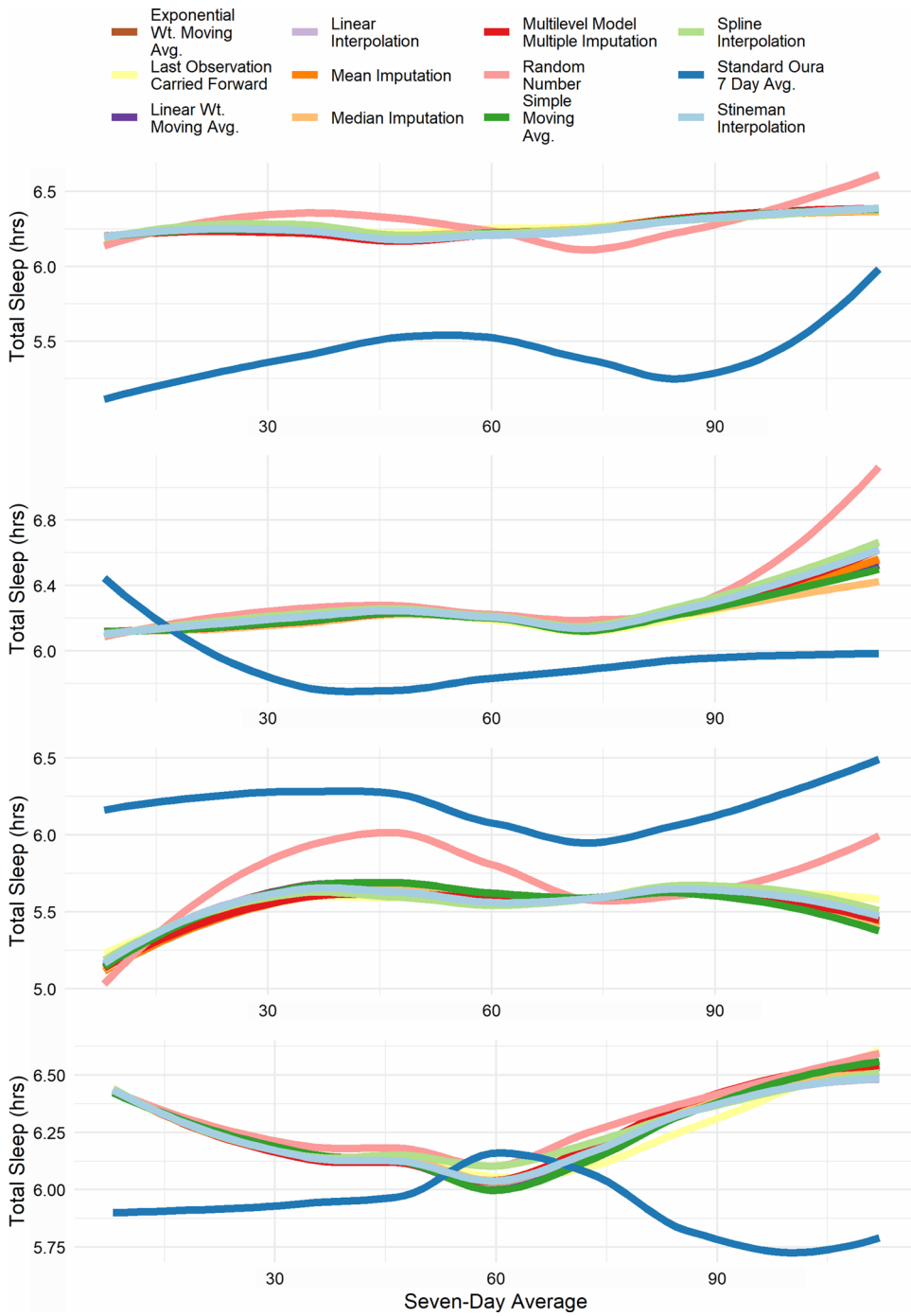

Fig. 5 Four representative example athletes showing how a 7-day average for total sleep using pairwise deletion (*dark blue line*) is qualitatively different from the imputation methods. The *pink line*, using random number imputation, deviates from many of the imputation methods as well. The images for all Athletes are available as Supplementary Information

Practitioners and analysts will need to review their own data for missingness and explore potential patterns of missingness in their data to determine if the MCAR assumption is viable. If the MCAR assumption is violated, as we see in this cohort of data from Collegiate American Football, steps need to be taken to preserve important characteristics of the data in light of this non-ignorable missingness.

The common use of data for sport science practitioners is for real-time decision making; however, the vast majority of research into missing data, imputation techniques, and analysis focuses on fully collected data. While this is the first paper to address the topic of real-time imputation in sports medicine or sport science, others have recently addressed this topic in the field of cardiology [27, 28]. They tested different imputation methods using simulated data, including mean imputation and multilevel model multiple imputation (which they called joint modeling imputation); while they find that multilevel model multiple imputation outperforms mean imputation, their missing data meets the MCAR

criteria from the onset so the problem set is different from what one would likely see in the real world. The finding that multiple imputation-style methods perform better than more simplistic imputation methods is common across academic research [27–31].

Since MCAR is known to be violated, it is expected that deletion would provide a biased estimate of the characteristic of interest [7]. As can be seen in Fig. 5, all of the imputation methods seem to retain similar data characteristics to each other, especially when contrasted with pairwise deletion. While we know that the pairwise deletion estimates for the individual athlete are 'wrong', we do not know 'how right' the imputation estimates are, since we do not know the underlying data-generating mechanism. This is a known limitation of the current work and the analysis in Fig. 4 is comparing each imputation type against itself, not a known ground truth. The empirical work in the area of data imputation indicates that an appropriately specified multiple imputation model should produce the best estimates in most cases [27–31], as they have the ability to preserve the relationships among the observed data points. Some of the univariate imputation methods also make implicit assumptions that are often unreasonable. For example, last-observation-carried-forward assumes a perfect relationship between the last observation and those following it without considering the overall patient/athlete trajectory. However, in our current low-information example with the American Football team, univariate moving averages were shown to be relatively stable estimates across time when data are iteratively added across the season. Therefore, the current suggested course of action for dashboards in sport science is to not assume that data are MCAR unless proven otherwise and also consider an appropriate univariate imputation technique (such as exponentially weighted moving averages) or multiple imputation when possible. End-users of sport science dashboards need to confer with the creators of their content to determine if the data aggregations and summaries are valid representations of both their team and individual athletes.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s40279-023-01829-w.

Data Availability Anonymized data and code can be provided upon written request that is approved by the West Virginia University Athletics Department.

Declarations

Conflict of Interest The author has no funding or competing interests to declare.

Ethical Approval All research analysis performed in this manuscript is approved by the WVU Institutional Review Board as a retrospective analysis of data with a waiver of informed consent in accordance with the Helsinki Declaration.

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